Chapter A

Duties under surveillance

Two versions are proposed for each chapter. The reference structure is: fluency of 6 points, exercise on 6 points and problem on 8 points.

Chapter 1 · Numerical sequences

Duty to Monitor 1 · Version A

Numerical sequences **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Fluency

[6 points]

1. Give the multiplier factor associated with an increase of 5 per cent.
2. Is continuation 12, 17, 22, 27, . . . arithmetic? Give common difference.
3. How many terms contain $u_4 + u_5 + \cdots + u_{18}$?
4. If $w_0 = 80$ and $w_{n+1} = 0{,}9w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Training 2 · Subscriptions

[6 points] A platform has 120,000 subscribers at the beginning of the year and earns 11,000 each Months. u_n the number of subscribers, in thousands, after n months.

1. Justify that (u_n) is arithmetic and specifies its common difference.
2. Express u_n according to the n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Training 3 · Placement and Objective

[8 points] An initial capital of 650,000 Ar increases by 5% per year. C_n the capital after n years.

1. Write C_{n+1} according to the $C_n$then C_n according to the n.
2. Calculate C_5, rounded to l ·ariary.
3. Determine to the calculator the first integer n for which C_n Exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Detailed correction · Professor edition

Scale. Ex. 1 : 6 points spread over fluency. Ex. 2 : 6 points. Ex. 3 : 8 points.
Elements of correction. The coefficient is 1 ,05. The first flash sequence is arithmetical of common difference 5 ; the sum contains 15 terms ; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 120 + 11n$, therefore $u_{12} = 252$; the amount claimed uses 12 $\dfrac{u_1 + u_{12}}{2}$ and cumulates monthly statements, not a number

For fiscal year 3, $C_n = 650\,000 \times \left(1 + \dfrac{5}{100}\right)^n$; control the threshold by two consecutive values.

Duty to Monitor 1 · Version B

Numerical sequences **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Fluency

[6 points]

1. Give the multiplier factor associated with an increase of 4 per cent.
2. Is continuation 12, 17, 22, 27, . . . arithmetic? Give common difference.
3. How many terms contain $u_4 + u_5 + \cdots + u_{18}$?
4. If $w_0 = 80$ and $w_{n+1} = 0{,}9w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Training 2 · Subscriptions

[6 points] A platform has 145,000 subscribers at the beginning of the year and earns 9,000 each. Months. u_n the number of subscribers, in thousands, after n months.

1. Justify that (u_n) is arithmetic and specifies its common difference.
2. Express u_n according to the n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Training 3 · Placement and Objective

[8 points] An initial capital of 720 000 Ar increases by 4 % per year. C_n the capital after n years.

1. Write C_{n+1} according to the $C_n$then C_n according to the n.
2. Calculate C_5, rounded to l ·ariary.
3. Determine to the calculator the first integer n for which C_n Exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Detailed correction · Professor edition

Scale. Ex. 1 : 6 points spread over fluency. Ex. 2 : 6 points. Ex. 3 : 8 points.

Elements of correction. The coefficient is 1 ,04. The first flash sequence is arithmetical of common difference 5; the sum contains 15 terms; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 145 + 9n$, therefore $u_{12} = 253$; the amount requested uses 12 $\dfrac{u_1 + u_{12}}{2}$ and collects monthly statements, not a number of people For fiscal year 3, $C_n = 720\,000 \times \left(1 + \dfrac{4}{100}\right)^n$; control the threshold by two values consecutive.

Chapter 2 · exponential functions

Must be monitored 2 · Version A

Exponential functions **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, represent, model, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Powers and Variations

[6 points]

1. Simplify $2^{4} \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^{x}}$.
2. Write 3^{-2} in fraction form.
3. Provide a sense of variation in $f(x) = 1{,}04^{x}$ and $g(x) = -2 \times 0{,}75^{x}$.
4. Explain why $a^{x} + a^{y} = a^{x+y}$ is generally false.

Training 2 · Attendance at a site

[6 points] At the launch, a site receives 2,400 visits per day.
We model by $V(t) = 2400 \times 1{,}04^{t}$.

1. Interpret the numbers 2400 and 1.04.
2. Calculate $V(8)$, rounded to one unit.
3. Determine the first integer number of weeks for which $V(t) > 3100$.

Training 3 · Average rate

[8 points] A company's turnover rose from 82,000 to 104,000 euros in four years.

1. Calculate the overall coefficient.
2. Write the average annual coefficient as a power.
3. Calculate the average annual rate, rounded to 0.01 per cent.
4. Explain why the overall rate is not simply divided by four.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** $2^{4} \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^{x}} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g
In the problem, calculate $V(8)$ then test two consecutive integers around the threshold.
For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; coefficients multiply.

Duty to Monitor 2 · Version B

Exponential functions **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, represent, model, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Powers and Variations

[6 points]

1. Simplify $2^{4} \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^{x}}$.
2. Write 3^{-2} in fraction form.
3. Provide a sense of variation in $f(x) = 1{,}03^{x}$ and $g(x) = -2 \times 0{,}75^{x}$.
4. Explain why $a^{x} + a^{y} = a^{x+y}$ is generally false.

Training 2 · Attendance at a site

[6 points] At launch, a site receives 1850 visits per day.
We model by $V(t) = 1850 \times 1{,}03^{t}$.

1. Interpret the numbers 1850 and 1.03.
2. Calculate $V(8)$, rounded to one unit.
3. Determine the first integer number of weeks for which $V(t) > 2250$.

Training 3 · Average rate

[8 points] A company's turnover rose from ·82 000 to ·98000 in four years.

1. Calculate the overall coefficient.
2. Write the average annual coefficient as a power.
3. Calculate the average annual rate, rounded to 0.01 per cent.
4. Explain why the overall rate is not simply divided by four.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** $2^{4} \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^{x}} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g In the problem, calculate $V(8)$ then test two consecutive integers around the threshold.

For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; coefficients multiply.

Chapter 3 · Decimal logarithm function

Duty to Monitor 3 · Version A

Decimal logarithm **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, solve, common difference / ratio, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Knowledge and calculations

[6 points]

1. Calculate $\log(1000)$, $\log(0{,}01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, develop $\log\left(\dfrac{100x^3}{y}\right)$.
3. Resolve $10^{t} = 28000$ and give t One hundredth.

Training 2 · Growth target

[6 points] A value is multiplied each year by 1.05.
double.

1. Translate the problem with an equation.
2. Solve this equation using the decimal logarithm.
3. Give the first integer number of years to reach or exceed the double.
4. Check with the two consecutive useful powers.

Training 3 · Sound level

[8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$ In decibels.

1. Calculate L where $I = 10^6 I_0$.
2. For $L = 84$, express and calculate the report $\dfrac{I}{I_0}$.
3. Compare intensities corresponding to 84 dB and 74 dB.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** The usual values are 3, $-2, -4$. Development $2 + 3\log x - \log y$. $t = \log(28000)$. For doubling, $x = \dfrac{\log 2}{\log(1{,}05)}$ then we take the integer thing. and we control. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; A difference of 10 dB corresponds to an intensity multiplied by 10.

Duty to Monitor 3 · Version B

Decimal logarithm **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, solve, common difference / ratio, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Knowledge and calculations

[6 points]

1. Calculate $\log(1000)$, $\log(0{,}01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, develop $\log\left(\dfrac{100x^3}{y}\right)$.
3. Resolve $10^{t} = 32000$ and give t One hundredth.

Training 2 · Growth target

[6 points] A value is multiplied each year by 1.07.
double.

1. Translate the problem with an equation.
2. Solve this equation using the decimal logarithm.
3. Give the first integer number of years to reach or exceed the double.
4. Check with the two consecutive useful powers.

Training 3 · Sound level

[8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$ In decibels.

1. Calculate L where $I = 10^6 I_0$.
2. For $L = 78$, express and calculate the report $\dfrac{I}{I_0}$.
3. Compare intensities corresponding to 78 dB and 68 dB.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** The usual values are 3, $-2, -4$. Development $2 + 3\log x - \log y$. $t = \log(32000)$. For doubling, $x = \dfrac{\log 2}{\log(1{,}07)}$ then we take the integer thing. and we control. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; A difference of 10 dB corresponds to an intensity multiplied by 10.

Chapter 4 · Reciprocal function

Duty to Monitor 4 · Version A

Reciprocal function **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, common difference / ratio, represent, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Reciprocal function

[6 points] It is considered $f(x) = 2 + \dfrac{16}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its sense of variation over each interval of the domain.

Training 2 · Signboard

[6 points] Solve $\dfrac{(x-4)(x+2)}{x} \geq 0$. The table shall show the zeros and the value Not allowed.

Training 3 · Average Cost

[8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{16}{x} + 12$.

1. Calculate $C'(x)$ and factor his numerator.
2. Draw up the table of changes in C on $]0, +\infty[$.
3. Determine the value of x which minimizes the average cost and calculates this minimum.
4. Make a coherent outline of the curve.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{16}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$.

For the quotient, order -2, 0, 4 and exclude 0. $C'(x) = 1 - \dfrac{16}{x^2} = \dfrac{(x-4)(x+4)}{x^2}$;
minimum in $x = 4$ of value 20.

Duty to Monitor 4 · Version B

Reciprocal function **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: calculate, common difference / ratio, represent, communicate

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Reciprocal function

[6 points] It is considered $f(x) = 2 + \dfrac{9}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its sense of variation over each interval of the domain.

Training 2 · Signboard

[6 points] Solve $\dfrac{(x-3)(x+2)}{x} \geq 0$. The table shall show the zeros and the value
Not allowed.

Training 3 · Average Cost

[8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{9}{x} + 12$.

1. Calculate $C'(x)$ and factor his numerator.
2. Draw up the table of changes in C on $]0, +\infty[$.
3. Determine the value of x which minimizes the average cost and calculates this minimum.
4. Make a coherent outline of the curve.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{9}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$.
For the quotient, order -2, 0, 3 and exclude 0. $C'(x) = 1 - \dfrac{9}{x^2} = \dfrac{(x-3)(x+3)}{x^2}$;
minimum in $x = 3$ of value 18.

Chapter 5 · Statistics with two quantitative variables

Duty to Monitor 5 · Version A

Two-variate statistics **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: represent, model, calculate, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Build and describe

[6 points] Couples are observed $(1;8)$, $(2;11)$, $(3;13)$, $(4;17)$, $(5;19)$ and $(6;23)$.

1. Represent the cloud in a correctly leged coordinate system.
2. Calculate the coordinates of the mean point.
3. Describe the trend and dispersal without concluding a causality.

Training 2 · Applying an adjustment

[6 points] A right of adjustment is $y = 3{,}2x + 12$, prepared for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and qualify the estimate.
2. Estimate x where $y = 60$.
3. Estimate y for $x = 40$, then explain why this result should be considered with caution.

Training 3 · Change of variable

[8 points] Set $z = \log(y)$ and the adjustment of (x, z) is $z = 0{,}06x + 1{,}1$.

1. Express y according to the x.
2. Estimate y for $x = 8$.
3. Explain the value of variable change and an extrapolation limit.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ From six values. With the right, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a remote extrapolation. last model is $y = 10^{0{,}06x+1{,}1}$.

Duty to Monitor 5 · Version B

Two-variate statistics **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: represent, model, calculate, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Build and describe

[6 points] Couples are observed $(1;8)$, $(2;11)$, $(3;13)$, $(4;17)$, $(5;22)$ and $(6;23)$.

1. Represent the cloud in a correctly leged coordinate system.
2. Calculate the coordinates of the mean point.
3. Describe the trend and dispersal without concluding a causality.

Training 2 · Applying an adjustment

[6 points] A right of adjustment is $y = 2{,}4x + 18$, prepared for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and qualify the estimate.
2. Estimate x where $y = 60$.
3. Estimate y for $x = 40$, then explain why this result should be considered with caution.

Training 3 · Change of variable

[8 points] Set $z = \log(y)$ and the adjustment of (x, z) is $z = 0{,}06x + 1{,}1$.

1. Express y according to the x.
2. Estimate y for $x = 8$.
3. Explain the value of variable change and an extrapolation limit.

Detailed correction · Professor edition

Scale: $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ From six values. With the right, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a remote extrapolation. last model is $y = 10^{0{,}06x+1{,}1}$.

Chapter 6 · Conditional Probabilities

Duty to Monitor 6 · Version A

Conditional probabilities **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Weighted shaft

[6 points] A ratio of 0.35 orders is express. The delay rate is 0.05 among the express commands and 0.12 among others.

1. Defining Events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ by the formula of total probabilities.
4. Calculate $P_R(E)$ and interpret.

Training 2 · Independence

[6 points] In another study, $P(A) = 0{,}4$, $P(B) = 0.25$ and $P(A \cap B) = 0{,}10$.

1. Calculate $P(A)P(B)$.
2. Are events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Training 3 · Count table

[8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non premium customers Build the complete table and then calculate the probability of being premium knowing that the client has terminated.

Detailed correction · Professor edition

Scale: $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For Independence, compare intersection to product 0 ,10. In the table there are 168 terminations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Duty to Monitor 6 · Version B

Conditional probabilities **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Weighted shaft

[6 points] A ratio of 0.45 orders is express. The delay rate is 0.08 among the express controls and 0.15 among others.

1. Defining Events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ by the formula of total probabilities.
4. Calculate $P_R(E)$ and interpret.

Training 2 · Independence

[6 points] In another study, $P(A) = 0{,}4$, $P(B) = 0.25$ and $P(A \cap B) = 0{,}14$.

1. Calculate $P(A)P(B)$.
2. Are events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Training 3 · Count table

[8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non premium customers Build the complete table and then calculate the probability of being premium knowing that the client has terminated.

Detailed correction · Professor edition

Scale: $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For Independence, compare intersection to product 0 ,10. In the table there are 168 terminations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Chapter 7 · Discreet random variables and binomial distribution

Duty to Monitor 7 · Version A

Random variables and binomial distribution **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Expectation of a game

[6 points] A game costs 3 euros. The gross gain is 0, 5 or 15 euros with probability 0 ,65, 0,25 and 0,10.

1. Build the law of net gain G.
2. Calculate $E(G)$.
3. Is the game favorable to the player? Interpret without saying what will happen during a game.

Training 2 · Binomial Distribution

[6 points] A client responds favourably with probability 0.30. An advisor contacts Independently 10 customers. X the number of favourable replies.

1. Justifying the law of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and Interpret $E(X)$.

Training 3 · Triangle de Pascal

[8 points]

1. Write the line $n = 6$ of Pascal's triangle.
2. Read $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Detailed correction · Professor edition

Scale: $6 + 8 + 6$. **Solution:** Net gain is $-3, 2, 12$; $E(G) = -1{,}95 + 0{,}50 + 1{,}20 = -0{,}25$ euro. $X \sim \mathcal{B}(10, 0{,}30)$; apply the formula, use 1 $- P(X = 0)$ for "at least one", and $E(X) = np$. The line $n = 6$ is 1, 6, 15, 20, 15, 6, 1.

Duty to Monitor 7 · Version B

Random variables and binomial distribution **Duration: 1 hour Note: /20**
Name: _____ Class: _____

Competences: model, calculate, common difference / ratio, interpret

The calculator is allowed unless otherwise indicated. All procedures must be legible and the results contextualized.

Training 1 · Expectation of a game

[6 points] A game costs 3 euros. The gross gain is 0, 5 or 15 euros with probability 0 ,65, 0,25 and 0,10.

1. Build the law of net gain G.
2. Calculate $E(G)$.
3. Is the game favorable to the player? Interpret without saying what will happen during a game.

Training 2 · Binomial Distribution

[6 points] A client responds favourably with probability 0.25. An advisor contacts Independently 8 customers. X the number of favourable replies.

1. Justifying the law of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and Interpret $E(X)$.

Training 3 · Triangle de Pascal

[8 points]

1. Write the line $n = 6$ of Pascal's triangle.
2. Read $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Detailed correction · Professor edition

Scale: $6 + 8 + 6$. **Solution:** Net gain is $-3, 2, 12$; $E(G) = -1{,}95 + 0{,}50 + 1{,}20 = -0{,}25$ euro. $X \sim \mathcal{B}(8, 0{,}25)$; apply the formula, use 1 $- P(X = 0)$ for "at least one", and $E(X) = np$. The line $n = 6$ is 1, 6, 15, 20, 15, 6, 1.